Agentic AI

What is it?

The Evolution of AI Systems

- Key points:
  - **Rule-based systems** – early AI with predefined rules.
  - **Machine Learning systems** – models that learn patterns from data.
  - **Generative AI / LLMs** – systems that generate text, code, images.
  - **Agentic AI** – systems that can **plan, reason, and take actions autonomously**.
- Example: Chatbots answer questions,
  - Agentic AI **solves tasks end-to-end**.

What is Agentic AI?

Agentic AI refers to AI systems that can **perceive goals, plan actions, use tools, and execute tasks autonomously with minimal human intervention.**

- Key characteristics:
  - Goal-driven behaviour, Autonomy, Planning and reasoning, Interaction with tools and environments, Continuous learning or feedback loops
- Example: Instead of answering *"How to plan a trip?"*
  An agent **books flights**, **reserves hotels**, **and generates the itinerary**.

Core Components of Agentic AI: 1. Perception

- Understanding inputs from: text, images, APIs, databases
  - Sensors & Input Interfaces: Capture data from the environment (text, voice, images, structured data).
  - Natural Language Understanding (NLU): Enables comprehension of human instructions and context.
  - Data Integration: Pulls information from APIs, databases, and external systems.

Core Components of Agentic AI: 2. Reasoning

- The AI decides **what steps are needed** to achieve a goal.
- Techniques include:
  - chain-of-thought reasoning
  - planning algorithms
  - task decomposition

Core Components of Agentic AI: 2. Reasoning

- Reasoning & Decision Making
  - Conditional Logic (if-then rules)
  - Heurestics (search algorithm)
  - ReAct (Reason+Act)
  - ReWOO (Reason WithOut Obesrevation)
  - Self-reflection
  - Multiagent Reasoning
  - Challenges: Computational Complexity, Interpretability (explainability, transparency), Scalability

Core Components of Agentic AI: 3. Planning

- Breaking a large task into **subtasks**.
- Example:
  Goal: Organize a conference
- Plan:
  - Identify venue
  - Contact speakers
  - Schedule sessions
  - Manage registrations

Core Components of Agentic AI: 4. Action / Tool Use

- Agents can interact with:
  - APIs
  - databases
  - software tools
  - external services
- Example:
  - Send emails
  - Query databases
  - Run code

Core Components of Agentic AI:5. Memory

- Agents store information such as:
  - previous interactions
  - intermediate results
  - learned preferences
- Two types:
  - short-term memory (session context):Stores immediate context for ongoing tasks.

  - long-term memory (knowledge storage):Retains knowledge, preferences, and historical interactions.

Vector Databases / Knowledge Graphs: Organize information for retrieval and reasoning.

Core Components of Agentic AI: 5. Memory

- Memory
  - Short-Term Memory
    - Context Window
    - State?
  - Long-Term Memory
    - Episodic Memory (past experiences)
    - Semantic Memory (factual knowledge)
    - Procedural Memory (skills, rules, learned behaviour)

Core Components of Agentic AI:5. Memory

In frameworks like LangChain, these are implemented as:

- ConversationBufferMemory (short-term)
- VectorStoreRetrieverMemory or Knowledge Graph Memory (long-term)

Agent Architecture

- Explain how agents are structured.
- Common architectures:
  - **Single-agent systems**
  - **Multi-agent systems**
- In **multi-agent systems**, agents collaborate: Examples:
  - Research agent
  - Planning agent
  - Execution agent
  - Evaluation agent

- This mirrors **human teams working together**.

How Agentic AI Works (Workflow)

- A simple workflow helps audiences understand the process.
- Typical cycle:
  - **Receive a goal**
  - **Analyze the task**
  - **Create a plan**
  - **Execute actions**
  - **Evaluate results**
  - **Refine and iterate**
- This is often called the **agent loop**.

Technologies Behind Agentic AI

- Explain the enabling technologies:
  - **Large Language Models (LLMs)**
  - **Reinforcement learning**
  - **Knowledge graphs**
  - **Vector databases**
  - **Tool APIs**
  - **Workflow orchestration frameworks**
- Examples of **frameworks:**
  - LangChain
  - AutoGPT
  - CrewAI
  - Microsoft AutoGen

https://www.youtube.com/watch?v=1bUy-1hGZpI

Related Topics

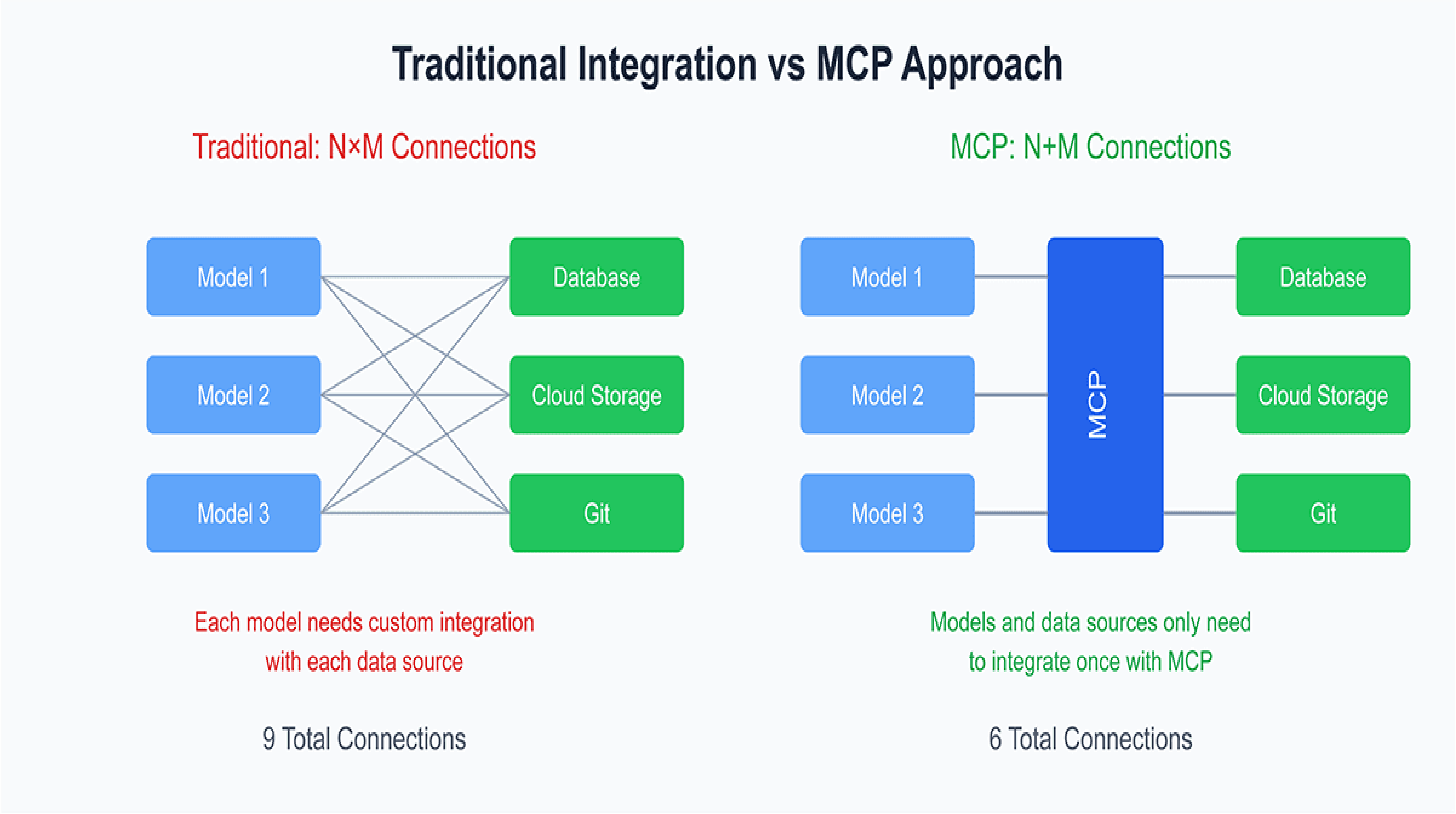
What is RAG? Retrieval-Augmented Generation

- **RAG =** It is a method that allows an AI model to **look up external information before answering** instead of relying only on what it memorized during training.
- **Simple idea**: Instead of guessing, the AI **searches first, then answers**.
- **How RAG works (step-by-step)**
- **User asks a question**: *"What are the policies for population data privacy?"*
- **Retriever searches a knowledge base**: This could be:
  - PDFs
  - Policies
  - Databases
  - Websites
  - Internal documents
- **Relevant documents are found**
- **Generator (LLM) reads them**
- **AI produces an answer based on the retrieved data**

https://www.youtube.com/watch?v=_HQ2H_0Ayy0

Why RAG is Important

- Without RAG, AI models:
- May hallucinate
  May use outdated knowledge
  Cannot access private/internal data
- With RAG:
- Uses **trusted sources**
  Uses **latest data**
  Works with **organization documents**
  Improves **accuracy and trust**



Traditional Integration vs MCP Approach
Traditional: N×M Connections
Model 1
Model 2
Model 3
Database
Cloud Storage
Git
Each model needs custom integration
with each data source
9 Total Connections
MCP: N+M Connections
Model 1
Model 2
Model 3
MCP
Database
Cloud Storage
Git
Models and data sources only need
to integrate once with MCP
6 Total Connections

Model Context Protocol (MCP)

- the **Model Context Protocol (MCP)** is very much part of the agentic AI ecosystem. It’s not a core reasoning or memory component itself, but rather a **standardized protocol** that enables AI models (especially LLM-based agents) to interact with external tools, data sources, and environments in a consistent way.
- MCP sits between the Reasoning/Memory layers and the Tools/Action layer.

- https://www.youtube.com/watch?v=E2DEHOEbzks

How RAG Fits with Agentic AI

- **RAG is a tool inside Agentic AI.**
- Agentic AI **uses RAG when it needs reliable information.**
- Think of:
- **Agent = Worker**
  RAG = Library
- The worker (agent) goes to the library (RAG) when it needs knowledge.

What is Agentic AI?

Advantages

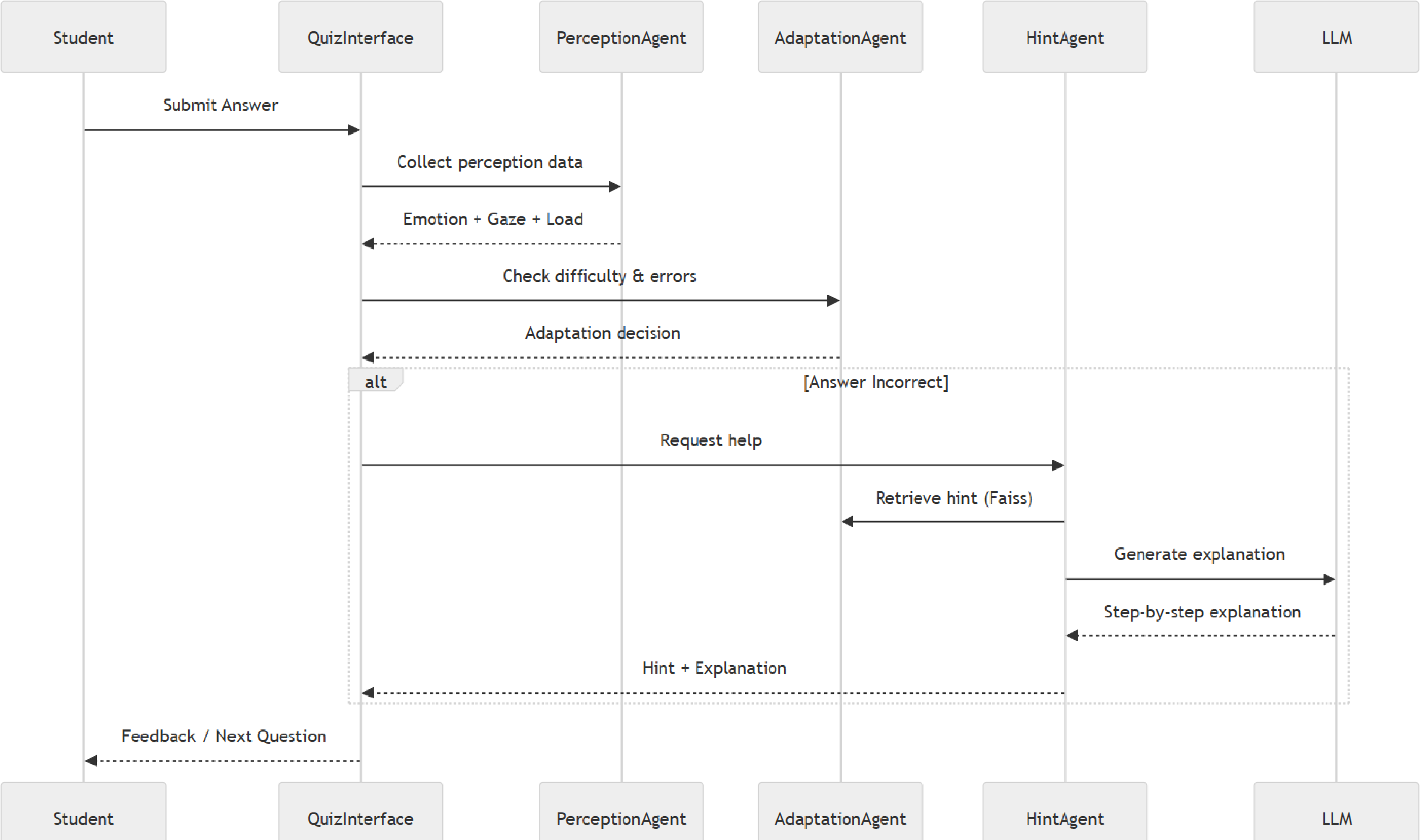
- Autonomy (range of autonomy)
- Proactive (respond to context)
- Specialised (multi-agent architectures)
- Adaptable (adjust behaviour based on feedback)
- Intuitive (natural language interface)

Disadvantages

- Agentic AI magnifies AI risk
- AI Safety (AI Alignment)
  - **Bias & Fairness**, Privacy, Loss of Control, Malicious **Misuse**, Existential Risks (AGI, ASI)
- AI Security
  - Data Security Risks, AI Model Risks, **Adversarial Attacks (Prompt Injection)**, Input Manipulation Attacks (Data Poisoning), Supply Chain Attacks, AI Model Drift & Decay, Regulatory Compliance

7. Real-World Applications

- **Business**
- automated research, market analysis, customer support
- **Government / Bureaucracy**
- policy document analysis, automated reporting, decision support systems, citizen service automation
- **Education**
- AI teaching assistants, personalized learning agents, research assistants for students
- **Software Engineering**
- autonomous coding agents, debugging systems, software design assistants


Student
QuizInterface
PerceptionAgent
AdaptationAgent
HintAgent
LLM
Submit Answer
Collect perception data
Emotion + Gaze + Load
Check difficulty & errors
Adaptation decision
alt
[Answer Incorrect]
Request help
Retrieve hint (Faiss)
Generate explanation
Step-by-step explanation
Hint + Explanation
Feedback / Next Question
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